

Images and Transformations

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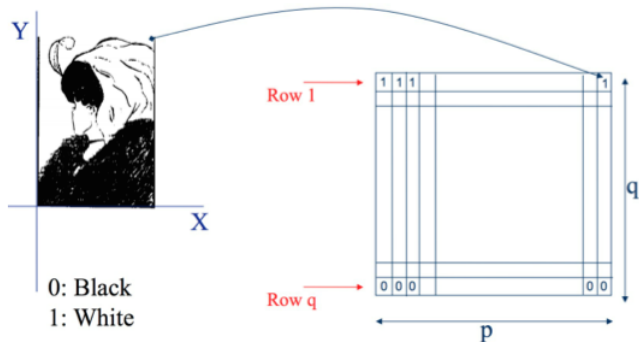
- Images, Sampling and Quantization
- 2D Transformation

Types of Images

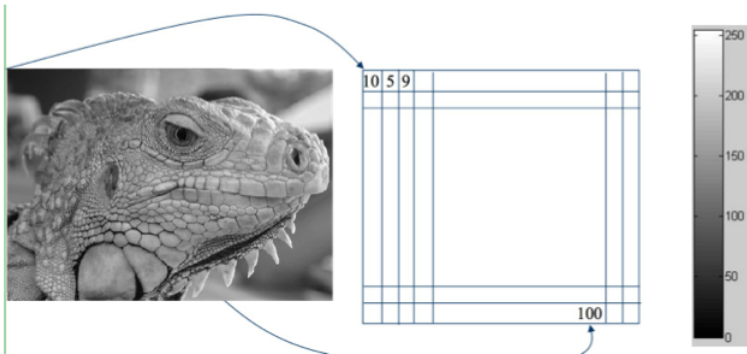
■ Types of Images: Binary, GrayScale and Color Image



Binary Image Representation



Grayscale Image Representation



Color Image Representation



B channel



G channel



R channel

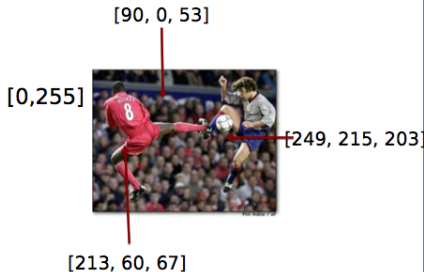
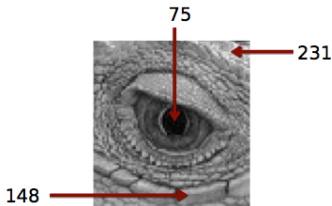
Digital Images are sampled

- What happens when we zoom into the images we capture?
- Resolution: is a sampling parameter, defined in dots per inch (DPI) or equivalent measures of spatial pixel density



Images are Sampled and Quantized

- An image contains discrete number of pixels
 - Pixel value:
 - "grayscale" (or "intensity"): $[0, 255]$
 - "color" - RGB = $[R, G, B]$



2D Transformation

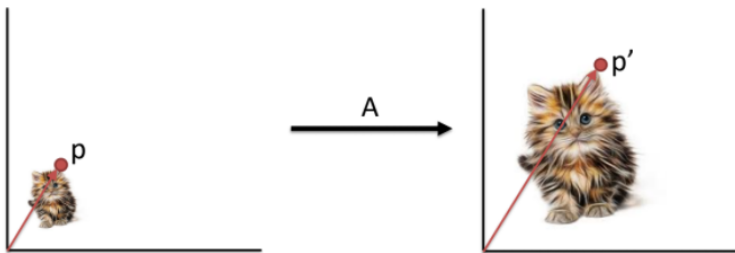
- Transformation Matrices
- Homogeneous coordinates
- Translation

- Matrices can be used to transform vectors in useful ways, through multiplication: $x' = Ax$
- Simplest is scaling:

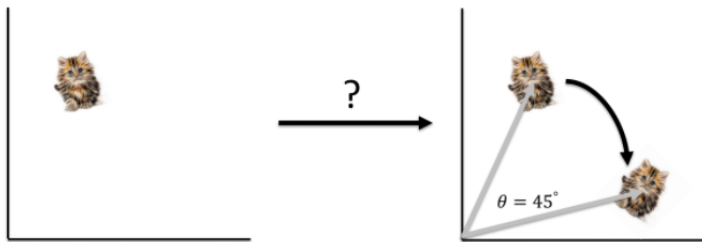
$$\begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} s_x x \\ s_y y \end{bmatrix}$$

Transformation

$$\begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} s_x x \\ s_y y \end{bmatrix}$$

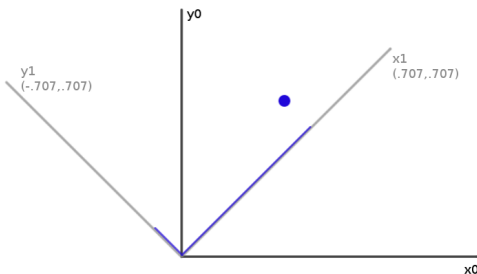


Rotation



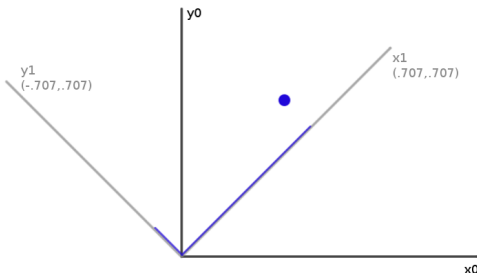
Rotation

- How can you convert a vector represented in frame “0” to a new, rotated coordinate frame “1”?
- Remember what a vector is: [component in direction of the frame’s x axis, component in direction of y axis]



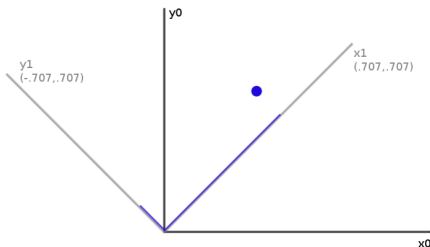
Rotation

- So to rotate it we must produce this vector: [component in direction of new x axis, component in direction of new y axis]
- We can do this easily with dot products!
- New x coordinate is [original vector] dot [the new x axis]
- New y coordinate is [original vector] dot [the new y axis]



Rotation

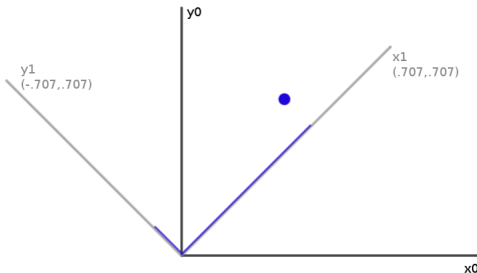
- Insight: this is what happens in a matrix * vector multiplication
 - Result x coordinate is [original vector] · [matrix row 1]
 - So matrix multiplication can rotate a vector p :



$$R \times p = \text{rotated } p'$$
$$R = \begin{bmatrix} .707 & .707 \\ -.707 & .707 \end{bmatrix}$$
$$p = \begin{bmatrix} px \\ py \end{bmatrix}$$
$$p' = \begin{bmatrix} px' \\ py' \end{bmatrix}$$

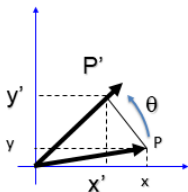
Rotation

- Suppose we express a point in a coordinate system which is rotated left
- If we use the result in the same coordinate system, we have rotated the point right
 - Thus, rotation matrices can be used to rotate vectors. We'll usually think of them in that sense— as operators to rotate vectors



2D Rotation Matrix Formula

Counter-clockwise rotation by an angle θ



$$x' = \cos \theta \, x - \sin \theta \, y$$

$$y' = \sin \theta \, x + \cos \theta \, y$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\mathbf{P}' = \mathbf{R} \, \mathbf{P}$$

Transformation Matrices

- Multiple transformation matrices can be used to transform a point: $p' = R_2 R_1 S p$
- The effect of this is to apply their transformations one after the other, from right to left.
- In the example above, the result is $(R_2(R_1(Sp)))$
- The result is exactly the same if we multiply the matrices first, to form a single transformation matrix: $p' = (R_2 R_1 S) p$